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Studierichting: Informatica
Oefeningenreeks 4A nr. 40.9

$$\int \frac{dx}{\sqrt{5-2x^2}}$$
$$= \int \frac{dx}{\sqrt{(\sqrt{5})^2 - (\sqrt{2}x)^2}}$$
Stel $\sqrt{2}x = y \Rightarrow \sqrt{2}dx = dy \Leftrightarrow dx = \frac{dy}{\sqrt{2}}$
$$I = \frac{1}{\sqrt{2}} \int \frac{dy}{\sqrt{(\sqrt{5})^2 - y^2}}$$
$$= \frac{1}{\sqrt{2}} \int \frac{dy}{\sqrt{5-y^2}}$$
$$= \frac{1}{\sqrt{2}} \cdot \frac{\mathrm{Bgsin}\left(\frac{y}{\sqrt{5}}\right)}{1} + C$$
$$= \frac{\mathrm{Bgsin}\left(\frac{y}{\sqrt{5}}\right)}{\sqrt{2}} + C$$
$$= \frac{\mathrm{Bgsin}\left(\frac{\sqrt{2}x}{\sqrt{5}}\right)}{\sqrt{2}} + C$$

References